NeuroFlow Hemodynamic Parameters

CFD & Morphological Metrics for Vascular Risk Assessment

1. Governing Physics: Blood Velocity & Pressure

In Computational Fluid Dynamics (CFD), blood velocity ($\mathbf{u}$) and pressure (p) fields are solved simultaneously using the incompressible **Navier–Stokes equations**.

Governing Continuum Equations:

$$\textbf{Continuity Equation (Mass Conservation):} \quad \nabla \cdot \mathbf{u} = 0$$

$$\textbf{Momentum Equation (Navier–Stokes):} \quad \rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

- $\mathbf{u} = (u_x, u_y, u_z)$: Blood velocity vector field derived by the CFD solver
- p : Static blood pressure distribution computed simultaneously
- ρ : Blood density (typically $\approx 1050\ \mathrm{kg/m^3}$)
- μ : Dynamic viscosity of blood ($\approx 0.0035\ \mathrm{Pa \cdot s}$)
- $\mathbf{f}$: External body forces (frequently neglected in neurovascular flow)

2. Fundamental Wall Parameter: Wall Shear Stress (WSS)

Wall Shear Stress ($\boldsymbol{\tau}_w$) is the tangential frictional force exerted by flowing blood on the endothelial layer of the vessel wall. It is the fundamental boundary metric from which all pulsatile derived indices originate.

Wall Shear Stress Vector Formula:

$$\boldsymbol{\tau}_w = \mu \left(\frac{\partial \mathbf{u}}{\partial n} \right)_{\text{wall}}$$

Where μ is dynamic viscosity, n is the wall normal direction, and $\left(\frac{\partial \mathbf{u}}{\partial n}\right)_{wall}$ is the velocity gradient at the vessel wall interface.

3. Pulsatile & Derived Hemodynamic Indices

During cardiac pulsation, flow direction and magnitude change dynamically over each heartbeat duration T. The derived metrics below integrate pulsatile behavior across the cardiac cycle.

3.1 Time-Averaged Wall Shear Stress (TAWSS)

TAWSS averages the magnitude of the instantaneous wall shear stress vector over an entire cardiac cycle (T).

$$\text{TAWSS} = \frac{1}{T} \int_0^T |\boldsymbol{\tau}_w(t)|\, dt$$

- **Higher TAWSS:** Indicates stronger blood friction on the vessel wall (high velocity flow).
- **Lower TAWSS:** May indicate blood stagnation, endothelial dysfunction, wall degeneration, and aneurysm growth.

3.2 Oscillatory Shear Index (OSI)

OSI measures the degree of directional deviation and reversal of the WSS vector relative to the main flow direction.

$$\text{OSI} = \frac{1}{2}\left(1 - \frac{\left|\int_0^T \boldsymbol{\tau}_w(t)\,dt\right|}{\int_0^T |\boldsymbol{\tau}_w(t)|\,dt}\right) \qquad \text{Bounded Range: } 0 \le \text{OSI} \le 0.5$$

- $\text{OSI} = 0:$ Blood flows strictly in one direction throughout the entire cycle.
- $\text{OSI} = 0.5:$ Blood continuously changes direction with zero net forward shear.
- **High OSI Significance:** Pinpoints disturbed flow, recirculation zones, vortex formation, and rupture-prone aneurysm regions.

3.3 Relative Residence Time (RRT)

RRT estimates how long blood particles remain near the vessel wall by coupling low shear stress magnitude with directional oscillation.

$$\text{RRT} = \frac{1}{(1 - 2\cdot\text{OSI}) \times \text{TAWSS}} = \frac{1}{\frac{1}{T}\left|\int_0^T \boldsymbol{\tau}_w(t)\,dt\right|}$$

- **High RRT Meaning:** Indicates slow, stagnant blood flow near the vessel wall.
- **Pathological Implications:** Highly correlated with thrombus formation, platelet deposition, and aneurysm progression.

3.4 Endothelial Cell Activation Potential (ECAP)

ECAP quantifies localized susceptibility to endothelial damage by taking the ratio of flow oscillation to shear magnitude.

$$\text{ECAP} = \frac{\text{OSI}}{\text{TAWSS}} \qquad [\text{Pa}^{-1}]$$

Higher ECAP $\longrightarrow$ Higher Endothelial Damage $\longrightarrow$ Higher Rupture Risk

4. Summary of NeuroFlow CFD Parameters

NeuroFlow visualizes fundamental flow features (*Pressure, Velocity, Streamlines, WSS*) and automatically derives pulsatile hemodynamic indicators (*TAWSS, OSI, RRT, ECAP*).

Parameter	Formula / Field Source	Clinical & Biomechanical Significance
Pressure (p)	Navier–Stokes Momentum	Hydrostatic pressure distribution, transmural wall forces.
Velocity ($\mathbf{u}$)	Continuity & Momentum	Streamlines, vortex cores, dynamic flow patterns.
WSS ($\boldsymbol{\tau}_w$)	$\mu(\partial\mathbf{u}/\partial n)_{\text{wall}}$	Instantaneous endothelial friction vector field.
TAWSS	$\frac{1}{T}\int_0^T \lvert\boldsymbol{\tau}_w\rvert dt$	Average shear load. Low $\rightarrow$ stagnation & wall weakening.
OSI	$\frac{1}{2}\left(1 - \frac{\lvert\int \boldsymbol{\tau}_w\rvert}{\int \lvert\boldsymbol{\tau}_w\rvert}\right)$	Shear reversal ($0 \le \text{OSI} \le 0.5$). High $\rightarrow$ recirculation & vortex.
RRT	$[(1-2\text{OSI})\text{TAWSS}]^{-1}$	Residence time near wall. High $\rightarrow$ thrombus growth.
ECAP	OSI/TAWSS	Oscillatory-to-shear ratio. High $\rightarrow$ peak endothelial damage.